

Dylan McDermott

West Des Moines, IA
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EDUCATION

Iowa State University, Ames, IA	January 2022 - May 2024
<i>Master of Science, Meteorology</i>	GPA: 3.52/4.0
● Master's Thesis: <i>Responses of Mesoscale Convective Systems to Land-Use Change</i>	

Iowa State University, Ames, IA	August 2018 - May 2022
<i>Bachelor of Science, Meteorology</i>	
● Senior Thesis: <i>Extreme Convective Wind Magnitude and Duration in Iowa</i>	
● Dean's List honors	

SKILLS

● Advanced research and analytical experience	● Numerical weather modeling and simulation
● Strong technical, analytic, and problem-solving skills	● Ability and willingness to work in multiple areas of science
● Experienced in computational modeling	● RADAR interpretation and remote sensing
● Manipulation of large datasets	● Machine learning and AI model development
● Deep understanding of ML/AI	● Catastrophe (CAT) meteorology and insurance risk analysis
● Displayed strong oral and written communication skills	● NWP data processing and analysis
● Interdisciplinary scientific research	

SOFTWARE

● Python, MATLAB, C++, FORTRAN, SQL	● Linux, Windows, Mac, Jupyter, AWS
● ArcGIS Pro, Experience Builder, Enterprise	● GR2Analyst

WORK EXPERIENCE

<i>Meteorologist</i>	January 2025-Present
Property & Liability Resource Bureau, Remote	
● Serve as in-house meteorologist for the Weather/CAT team, supporting insurer members with weather-event verification, catastrophe analytics, and cause-of-loss data interpretation.	
● Build and maintain ArcGIS-based weather and catastrophe database used in claims support, including severe weather, precipitation, smoke, evacuation, earthquake, and power outage products.	
● Automate recurring data workflows with Python and ArcPy, improving operational consistency and reducing manual handling of daily weather datasets.	
● Advise member companies on proper weather-data use, source limitations, uncertainty, and transparent methodology for insurance-facing products.	
● Contributed to GIS-based catastrophe analytics work recognized with Esri's 2025 Significant Achievement in GIS Award.	

Research Associate (Contract)

June 2024-December 2024

Corteva Agriscience, Johnston, IA

- Contributed to an R&D proof-of-concept project developing novel in-canopy microclimate data collection methods, collaborating with engineers and agronomists to optimize sensor placement and data interpretation
- Designed and built the Corteva "Gold Standard" weather station and engineered an automated antenna-based data relay system to transmit field measurements to the main facility in real time
- Applied Python, R, and GIS tools to process and analyze large datasets across multiple concurrent research projects
- Worked across disciplines including data science, meteorology, and agricultural engineering to support team objectives and deliver research contributions

Graduate Research Assistant

May 2022-May 2024

Iowa State University, Ames, IA

- Performed advanced research, integrating knowledge from various scientific domains to conduct model simulations, perform data analysis, and deliver written and oral presentations of research findings.
- Utilized Python and MATLAB on a daily basis to efficiently manipulate complex datasets and conduct in-depth analysis of model outputs.
- Collaborated across diverse scientific disciplines, including meteorology, agronomy, engineering, hydrology, computer science, biology, and chemistry, gaining valuable experience in interdisciplinary research.

Undergraduate Research Assistant

May 2021-August 2021

Iowa State University, Ames, IA

- Engaged in advanced research in boundary layer meteorology under the guidance of Dr. Ian Williams at Iowa State University.
- Skillfully identified relevant cases, generated a diverse range of data visualizations and plots, and actively participated in drawing potential conclusions from the findings.

PERSONAL PROJECTS

AI-Powered Weather Intelligence Application

2025-Present

- Full stack weather intelligence platform integrating 200+ data streams into AI models for historical, current, and forecast analysis.
- AWS 10,000 Aldeas Competition Semi-Finalist

AWARDS

ESRI Significant Achievement in GIS (SAG) Award 2025

2025

- Recognized for innovative use of GIS technology in meteorological catastrophe analysis; selected among ~197 recipients from 100,000+ Esri clients worldwide